

Andrey Derzhavin

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SUMMARY

CPU simulation engineer with 5+ years at Huawei. Led a functional CPU simulator from prototype to production, designed a parallel microarchitectural exploration framework, and built loop-analysis tooling. Core expertise: C++, Python, CPU simulation (QEMU, gem5), microarchitecture, AArch64/RISC-V, LLVM, performance profiling. Co-created and taught the “CPU & OS Simulation” course at MIPT and ITMO. Focused on CPU simulation, low-level systems, performance engineering, and HW/SW co-design.

EXPERIENCE

Huawei Software Engineer	Jul 2021 — Present <i>Moscow, Russia</i>
- Lead a project on autonomous ELF benchmark generation from execution traces, managing tasks, writing code, and driving delivery. - Led development of a functional CPU simulator from prototype to production within one year (team of two); designed a DBT-like translation pipeline that cut translation time in half. - Researched microarchitectural design space exploration using low-order parameters. Designed a parallel, simulator-agnostic framework that identifies optimal microarchitectural configurations. - Built a static loop analysis tool using Boost Graph Library. Studied dynamic loop classification and quantified potential IPC improvement from enhanced loop termination prediction via ISA hints. - Mentored five interns over 3 years through code reviews, design discussions, task planning, and professional development.	

Acronis Intern	Jul 2020 — Nov 2020 <i>Remote</i>
Investigated right-context grammars for static analysis of C++ source code; proposed several novel applications.	

EDUCATION

Moscow Institute of Physics and Technology PhD student, Computer Science (expected 2028)	Dolgoprudny, Russia Sep 2025 — Present
Moscow Institute of Physics and Technology M.Sc. in applied math and physics, with honours	Dolgoprudny, Russia Sep 2023 — Jul 2025
Moscow Institute of Physics and Technology B.Sc. in applied math and physics, , with honours	Dolgoprudny, Russia Sep 2019 — Jul 2023
Selected Coursework: “Uses and Applications of C++ Language” (MIPT, 2020–2021), lecturer K. Vladimirov (Intel).	
Physics and Mathematics Lyceum No. 30	Saint-Petersburg, Russia Sep 2017 — Jun 2019

PROJECTS

JitResearch (github.com/ProteusLab/JitResearch)	Sep 2025 — Present
Comparative study of modern JIT frameworks (LLVM ORC, AsmJit and other JIT backends) for CPU functional simulation. Evaluated compilation latency, code quality, and integration complexity. Results presented at C++Russia 2026.	
leech-compiler (github.com/derzhavin3016/leech-compiler)	Sep 2023 — May 2024
Designed an SSA-based compiler IR for a custom VM. Implemented optimization passes (constant folding, inlining, peepholes, checks elimination).	

TALKS & TEACHING

JIT libraries for CPU simulation: difficulties of choice (cppconf.ru/talks/20010048/)	May 2026
Talk at C++Russia 2026. Discussed trade-offs among JIT libraries for functional CPU simulation.	
CPU & OS Simulation course (MIPT, ITMO) (proteuslab.github.io/SimCourse/)	Sep 2024 — Present
Co-developed and co-taught the course. Covered CPU architecture, simulation, and OS fundamentals.	

TECHNICAL SKILLS

- Programming:** C++, C, Python
- CPU Architecture:** microarchitecture, AArch64, RISC-V, x86-64
- Simulation & Emulation:** QEMU, gem5, functional CPU simulation
- Compilers & JIT:** LLVM, LLVM IR, JIT/DBT
- Performance:** profiling, benchmarking, performance analysis
- Libraries & Frameworks:** Boost, Google Test, NumPy
- Tools & Platforms:** CMake, git, Linux
- Languages:** Russian (native), English (C1)